

AI News Agent System

Multi-Agent News Generation Platform

Version 1.0.0

Generated using LangGraph, LangChain, and FastAPI

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1 Overview

The **AI News Agent System** is a multi-agent, LLM-powered platform that automatically:

- Researches a news topic from the web
- Writes a creative news article
- Edits and refines the article
- Saves the final output as a Markdown file
- Exposes REST APIs for UI and programmatic access

The system is built using:

- **LangGraph** for agent orchestration
- **LangChain** for LLM and tools
- **FastAPI** for API services

2 System Architecture

2.1 High-Level Flow

1. Supervisor agent evaluates workflow state
2. Researcher agent gathers factual information
3. Writer agent creates a news article
4. Editor agent improves clarity and tone
5. File Writer agent saves Markdown output

2.2 Agent Graph

The workflow is implemented using a LangGraph **StateGraph**:

- Nodes represent agents
- Conditional routing determines next execution step
- Workflow ends only after file persistence

3 Agents Description

3.1 Supervisor Agent

Role: Orchestrates the workflow **Responsibilities:**

- Decides which agent runs next
- Ensures all steps are completed
- Prevents premature termination

3.2 Researcher Agent

Role: Factual data collection **Tool Used:** Tavily Web Search **Outputs:**

- Research summary
- Reference URLs

3.3 Writer Agent

Role: Creative article generation **Outputs:**

- Headline
- Introduction
- Key developments
- Conclusion

3.4 Editor Agent

Role: Content refinement **Improvements:**

- Professional tone
- Readability
- Factual consistency

3.5 File Writer Agent

Role: Persistence **Outputs:**

- Markdown file (.md)
- File path returned to API

4 Workflow State

The system uses a shared state object:

```
class SupervisorState(MessagesState):  
    next_agent: str  
    topic: str  
    research_summary: str  
    references: list[str]  
    article: str  
    final_article: str  
    markdown_content: str  
    task_complete: bool  
    file_path: str
```

This state is passed between agents during execution.

5 API Design

5.1 Health Endpoints

Endpoint	Description
GET /	Basic health check
GET /health	Detailed health check

5.2 Article Generation

Endpoint	Description
POST /api/v1/articles/generate	Generate article synchronously
GET /api/v1/articles/status/job_id	Check job status
GET /api/v1/articles/list	List generated articles

5.3 Download Endpoints

Endpoint	Description
GET /api/v1/articles/download-markdown	Download latest article
GET /api/v1/articles/download/filename	Download specific file

6 Markdown File Structure

Each generated article is saved as:

```
generated_articles/  
topic_timestamp.md
```

The file includes:

- Metadata header
- Final edited article
- References section

7 Execution Modes

7.1 API Mode (Default)

```
python main.py --mode api
```

7.2 CLI Mode

```
python main.py --mode cli
```

CLI mode allows manual topic entry and direct output.

8 Error Handling

- All exceptions are captured per job
- Job status is marked as FAILED on errors
- API returns error messages safely

9 Future Enhancements

- Streaming token responses
- Multiple article styles
- PDF and HTML exports
- Agent performance analytics
- Authenticated multi-user support

10 Conclusion

The AI News Agent System demonstrates a scalable, production-ready multi-agent architecture for automated content generation using LLMs, LangGraph, and FastAPI.